Lab_10

German Vega

```r
pdb_data <- read.csv("pdb_stats.csv", row.names = 1, header = TRUE)
rownames(pdb_data)  # should now show molecule type names
```

```
[1] "Protein (only)"          "Protein/Oligosaccharide"
[3] "Protein/NA"              "Nucleic acid (only)"
[5] "Other"                   "Oligosaccharide (only)"
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

About 93.79%

```r
# Q1: X-ray and EM percentages
xray_total  <- sum(pdb_data$X.ray)
em_total    <- sum(pdb_data$EM)
grand_total <- sum(pdb_data$Total)

round((xray_total / grand_total) * 100, 2)                # X-ray %
```

```
[1] 80.95
```

```r
round((em_total   / grand_total) * 100, 2)                # EM %
```

```
[1] 12.84
```

```r
round(((xray_total + em_total) / grand_total) * 100, 2)    # Combined %
```

```
[1] 93.79
```

Q2: What proportion of structures in the PDB are protein?

About 97.91%

```
# Q2: Proportion that is protein
protein_total <- sum(pdb_data[c("Protein (only)",
                                "Protein/Oligosaccharide",
                                "Protein/NA"), "Total"])
round((protein_total / grand_total) * 100, 2)
```

```
[1] 97.91
```

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 1.327 Structures.

Figure 1: HIV-1-Pr

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

> Water molecules normally have 3 atoms (1 oxygen + 2 hydrogens). However, in X-ray crystallography, hydrogen atoms are invisible because they have only 1 electron and produce no detectable electron density at that resolution. So only the oxygen atom of each water molecule appears in the structure.

Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

> The critical conserved water molecule is HOH 308. You can find it in Mol* by toggling waters on and looking in the binding site between the two protein flaps and the drug indinavir. It is conserved because it appears in virtually every HIV-1 protease crystal structure.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.

This figure shows the two chains of HIV-1 protease (Chain A in blue, Chain B in red) along with the bound ligand indinavir (green).

Figure 2: HIV-1 Protease with Indinavir

Q7: [Optional] As you have hopefully observed HIV protease is a homodimer (i.e. it is composed of two identical chains). With the aid of the graphic display can you identify secondary structure elements that are likely to only form in the dimer rather than the monomer?

The most notable secondary structure element that only forms in the dimer is the

4-stranded -sheet at the dimer interface, formed by the interdigitation of the N and C termini of both chains (residues 1–4 and 96–99). Neither monomer alone can form this sheet — it requires contributions from both chains, making it a true dimer-only structure that is critical for holding the two chains together and maintaining the overall stability of the enzyme.

```r
library(bio3d)
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
pdb
```

```

 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198 amino acid residues

Q8: Name one of the two non-protein residues?

HOH (water) or MK1 (indinavir)

Q9: How many protein chains are in this structure?

2 chains (A and B)

```
attributes(pdb)
```

```
$names
[1] "atom"   "xyz"    "seqres" "helix"  "sheet"  "calpha" "remark" "call"

$class
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

```
library(bio3dview)
library(NGLVieweR)

view.pdb(pdb) |>
  setSpin()
```

```
Warning in basename(x): expanded path length 136572 would be too long for
ATOM      1  N   PRO A   1      29.361  39.686   5.862  1.00 38.10           N
ATOM      2  CA  PRO A   1      30.307  38.663   5.319  1.00 40.62           C
ATOM      3  C   PRO A   1      29.760  38.071   4.022  1.00 42.64           C
ATOM      4  O   PRO A   1      28.600  38.302   3.676  1.00 43.40           O
```

```
ATOM      5  CB  PRO A   1      30.508  37.541   6.342  1.00 37.87           C
ATOM      6  CG  PRO A   1      29.296  37.591   7.162  1.00 38.40           C
ATOM      7  CD  PRO A   1      28.778  39.015   7.019  1.00 38.74           C
ATOM      8  N   GLN A   2      30.607  37.334   3.305  1.00 41.76           N
ATOM      9  CA  GLN A   2      30.158  36.492   2.199  1.00 41.30           C
ATOM     10  C   GLN A   2      30.298  35.041   2.643  1.00 41.38           C
ATOM     11  O   GLN A   2      31.401  34.494   2.763  1.00 43.09           O
ATOM     12  CB  GLN A   2      30.970  36.738   0.926  1.0 [... truncated]
```

```
Google Chrome was not found. Try setting the `CHROMOTE_CHROME` environment variable to the ex
based browser, such as Google Chrome, Chromium or Brave.
```

```r
# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
view.pdb(pdb, cols=c("navy","teal"),
         highlight = sele,
         highlight.style = "spacefill") |>
  setRock()
```

```
Warning in basename(x): expanded path length 136572 would be too long for
ATOM      1  N   PRO A   1      29.361  39.686   5.862  1.00 38.10           N
ATOM      2  CA  PRO A   1      30.307  38.663   5.319  1.00 40.62           C
ATOM      3  C   PRO A   1      29.760  38.071   4.022  1.00 42.64           C
ATOM      4  O   PRO A   1      28.600  38.302   3.676  1.00 43.40           O
ATOM      5  CB  PRO A   1      30.508  37.541   6.342  1.00 37.87           C
ATOM      6  CG  PRO A   1      29.296  37.591   7.162  1.00 38.40           C
ATOM      7  CD  PRO A   1      28.778  39.015   7.019  1.00 38.74           C
ATOM      8  N   GLN A   2      30.607  37.334   3.305  1.00 41.76           N
ATOM      9  CA  GLN A   2      30.158  36.492   2.199  1.00 41.30           C
ATOM     10  C   GLN A   2      30.298  35.041   2.643  1.00 41.38           C
ATOM     11  O   GLN A   2      31.401  34.494   2.763  1.00 43.09           O
ATOM     12  CB  GLN A   2      30.970  36.738   0.926  1.0 [... truncated]
```

```r
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

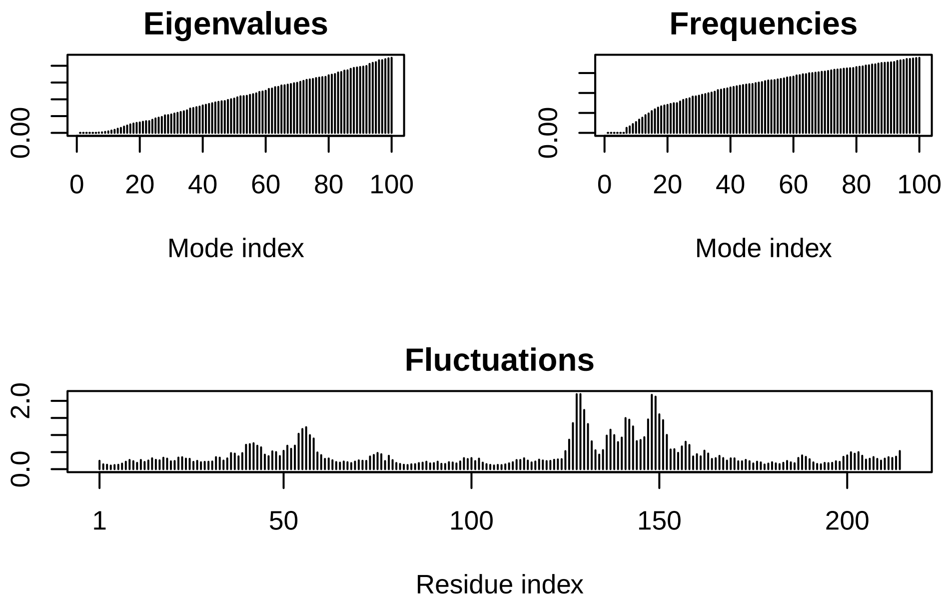
```r
adk
```

```
 Call:  read.pdb(file = "6s36")

   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
# Perform flexiblity prediction
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.01 seconds.
 Diagonalizing Hessian...   Done in 0.192 seconds.
```

```r
plot(m)
```




```
mktrj(m, file="adk_m7.pdb")
```

```
view.nma(m, pdb=adk)
```

```
Warning in basename(x): expanded path length 590110 would be too long for
MODEL        1
ATOM      2  CA  MET A   1     -19.326   9.682  18.832  1.00 36.10           C
ATOM     11  CA AARG A   2     -16.894   6.998  17.674  0.50 26.59           C
ATOM     32  CA  ILE A   3     -18.316   5.203  14.643  1.00 21.76           C
ATOM     40  CA  ILE A   4     -17.091   2.435  12.363  1.00 20.57           C
ATOM     48  CA  LEU A   5     -18.775   2.508   8.937  1.00 19.08           C
ATOM     56  CA  LEU A   6     -18.464  -0.698   6.902  1.00 24.62           C
ATOM     64  CA  GLY A   7     -19.758  -1.475   3.453  1.00 25.04           C
ATOM     68  CA  ALA A   8     -19.074  -3.493   0.350  1.00 26.93           C
ATOM     73  CA  PRO A   9     -17.159  -1.879  -2.535  1.00 27.11           C
ATOM     80  CA  GLY A  10     -19.711  -0.041  -4.657  1.00 30.50           C
ATOM     84  CA  ALA A  11     -22.527   0.023  -2.105  1.00 31.59           C
ATOM     89  CA  GLY A  12     -22.175   3.7 [... truncated]
```

Q10. Which of the packages above is found only on BioConductor and not CRAN?

> msa, it's installed via BiocManager::install() meaning it's only on BioConductor, not CRAN.

Q11. Which of the above packages is not found on BioConductor or CRAN?:

> bio3dview, it's installed via remotes::install_github() meaning it's only on GitHub, not on BioConductor or CRAN.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

> TRUE, the package can be used to install packages from GitHub

```r
library(bio3d)
aa <- get.seq("1ake_A")
```

```
Warning in get.seq("1ake_A"): Removing existing file: seqs.fasta
```

```
Fetching... Please wait. Done.
```

```r
aa
```

```
             1        .         .         .         .         .         60 
pdb|1AKE|A   MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
             1        .         .         .         .         .         60 

            61        .         .         .         .         .         120 
pdb|1AKE|A   DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
            61        .         .         .         .         .         120 

           121        .         .         .         .         .         180 
pdb|1AKE|A   VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
           121        .         .         .         .         .         180 

           181        .         .         .   214 
pdb|1AKE|A   YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
           181        .         .         .   214 

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

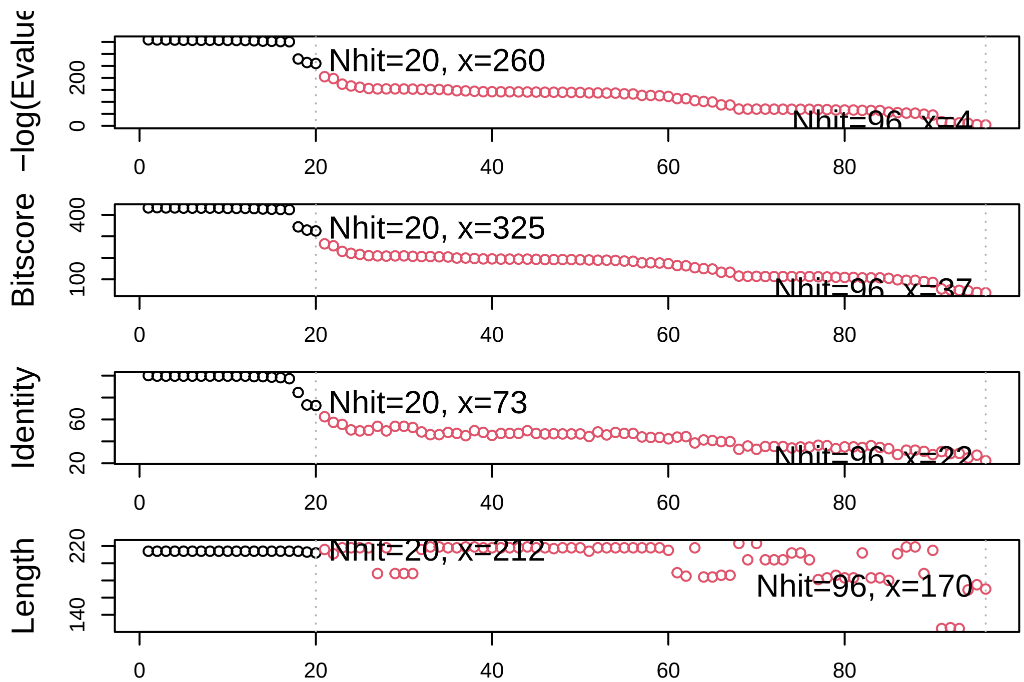
> There are 214 amino acids

```r
# Blast or hmmer search
b <- blast.pdb(aa)
```

```
 Searching ... please wait (updates every 5 seconds) RID = Z4RYMKX4016 
 .......................
 Reporting 96 hits
```

```r
# Plot a summary of search results
hits <- plot(b)
```

```
  * Possible cutoff values:    260 3
            Yielding Nhits:    20 96

  * Chosen cutoff value of:    260
            Yielding Nhits:    20
```




```r
# List out some 'top hits'
head(hits$pdb.id)
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A"
```

```r
its <- NULL
hits$pdb.id <- c('1AKE_A','6S36_A','6RZE_A','3HPR_A','1E4V_A','5EJE_A','1E4Y_A','3X2S_A','6HA
```

```r
# Download releated PDB files
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download
```

```
  |
  |                                                                      |   0%
  |
  |=====                                                                 |   8%
  |
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  |===========================================                           |  62%
  |
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  |======================================================                |  77%
  |
  |===========================================================           |  85%
  |
  |=================================================================     |  92%
  |
  |======================================================================| 100%
```

```r
# Align releated PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
```

```
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
....   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...

Extracting sequences

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/6S36_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/6RZE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdbs/split_chain/3HPR_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdbs/split_chain/5EJE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10   name: pdbs/split_chain/6HAM_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11   name: pdbs/split_chain/4K46_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13   name: pdbs/split_chain/4PZL_A.pdb
```

```r
library(bio3dview)

view.pdbs(pdbs)
```

```
Warning in basename(x): expanded path length 17340 would be too long for
```

```
ATOM      1  CA  MET A   1      26.091  52.849  39.889  1.00  0.00
ATOM      2  CA  ARG A   2      27.437  49.969  37.786  1.00  0.00
ATOM      3  CA  ILE A   3      24.961  47.988  35.671  1.00  0.00
ATOM      4  CA  ILE A   4      25.194  44.925  33.360  1.00  0.00
ATOM      5  CA  LEU A   5      22.428  44.503  30.712  1.00  0.00
ATOM      6  CA  LEU A   6      21.933  40.811  29.752  1.00  0.00
ATOM      7  CA  GLY A   7      19.604  39.512  26.973  1.00  0.00
ATOM      8  CA  ALA A   8      19.324  37.742  23.567  1.00  0.00
ATOM      9  CA  PRO A   9      20.108  39.651  20.294  1.00  0.00
ATOM     10  CA  GLY A  10      17.487  42.426  19.756  1.00  0.00
ATOM     11  CA  ALA A  11      15.960  42.201  23.284  1.00  0.00
ATOM     12  CA  GLY A  12      16.082  46.035  23.547  1.00 [... truncated]

Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      28.136  50.954  41.270  1.00  0.00
ATOM      2  CA  ARG A   2      29.515  48.252  38.977  1.00  0.00
ATOM      3  CA  ILE A   3      26.767  47.008  36.667  1.00  0.00
ATOM      4  CA  ILE A   4      26.590  44.476  33.853  1.00  0.00
ATOM      5  CA  LEU A   5      23.695  45.152  31.457  1.00  0.00
ATOM      6  CA  LEU A   6      22.730  42.233  29.206  1.00  0.00
ATOM      7  CA  GLY A   7      20.065  42.040  26.553  1.00  0.00
ATOM      8  CA  ALA A   8      19.166  40.405  23.284  1.00  0.00
ATOM      9  CA  PRO A   9      19.954  42.222  20.017  1.00  0.00
ATOM     10  CA  GLY A  10      17.033  44.540  19.315  1.00  0.00
ATOM     11  CA  ALA A  11      15.519  44.501  22.802  1.00  0.00
ATOM     12  CA  GLY A  12      16.563  48.111  23.486  1.00 [... truncated]
Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      28.136  50.954  41.270  1.00  0.00
ATOM      2  CA  ARG A   2      29.515  48.252  38.977  1.00  0.00
ATOM      3  CA  ILE A   3      26.767  47.008  36.667  1.00  0.00
ATOM      4  CA  ILE A   4      26.590  44.476  33.853  1.00  0.00
ATOM      5  CA  LEU A   5      23.695  45.152  31.457  1.00  0.00
ATOM      6  CA  LEU A   6      22.730  42.233  29.206  1.00  0.00
ATOM      7  CA  GLY A   7      20.065  42.040  26.553  1.00  0.00
ATOM      8  CA  ALA A   8      19.166  40.405  23.284  1.00  0.00
ATOM      9  CA  PRO A   9      19.954  42.222  20.017  1.00  0.00
ATOM     10  CA  GLY A  10      17.033  44.540  19.315  1.00  0.00
ATOM     11  CA  ALA A  11      15.519  44.501  22.802  1.00  0.00
ATOM     12  CA  GLY A  12      16.563  48.111  23.486  1.00 [... truncated]

Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      28.531  51.028  41.668  1.00  0.00
```

```
ATOM      2  CA  ARG A   2      29.513  48.225  39.235  1.00  0.00
ATOM      3  CA  ILE A   3      26.796  47.048  36.833  1.00  0.00
ATOM      4  CA  ILE A   4      26.648  44.525  34.012  1.00  0.00
ATOM      5  CA  LEU A   5      23.805  45.316  31.579  1.00  0.00
ATOM      6  CA  LEU A   6      22.777  42.431  29.312  1.00  0.00
ATOM      7  CA  GLY A   7      20.164  42.223  26.597  1.00  0.00
ATOM      8  CA  ALA A   8      19.213  40.704  23.247  1.00  0.00
ATOM      9  CA  PRO A   9      20.105  42.530  20.017  1.00  0.00
ATOM     10  CA  GLY A  10      17.199  44.849  19.279  1.00  0.00
ATOM     11  CA  ALA A  11      15.712  44.714  22.784  1.00  0.00
ATOM     12  CA  GLY A  12      16.768  48.307  23.422  1.00 [... truncated]
Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      28.531  51.028  41.668  1.00  0.00
ATOM      2  CA  ARG A   2      29.513  48.225  39.235  1.00  0.00
ATOM      3  CA  ILE A   3      26.796  47.048  36.833  1.00  0.00
ATOM      4  CA  ILE A   4      26.648  44.525  34.012  1.00  0.00
ATOM      5  CA  LEU A   5      23.805  45.316  31.579  1.00  0.00
ATOM      6  CA  LEU A   6      22.777  42.431  29.312  1.00  0.00
ATOM      7  CA  GLY A   7      20.164  42.223  26.597  1.00  0.00
ATOM      8  CA  ALA A   8      19.213  40.704  23.247  1.00  0.00
ATOM      9  CA  PRO A   9      20.105  42.530  20.017  1.00  0.00
ATOM     10  CA  GLY A  10      17.199  44.849  19.279  1.00  0.00
ATOM     11  CA  ALA A  11      15.712  44.714  22.784  1.00  0.00
ATOM     12  CA  GLY A  12      16.768  48.307  23.422  1.00 [... truncated]

Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      26.404  52.839  39.968  1.00  0.00
ATOM      2  CA  ARG A   2      27.389  49.887  37.790  1.00  0.00
ATOM      3  CA  ILE A   3      24.921  47.900  35.696  1.00  0.00
ATOM      4  CA  ILE A   4      25.122  44.958  33.303  1.00  0.00
ATOM      5  CA  LEU A   5      22.436  44.472  30.647  1.00  0.00
ATOM      6  CA  LEU A   6      21.779  40.833  29.752  1.00  0.00
ATOM      7  CA  GLY A   7      19.622  39.236  27.078  1.00  0.00
ATOM      8  CA  ALA A   8      19.187  37.774  23.609  1.00  0.00
ATOM      9  CA  PRO A   9      19.955  39.586  20.331  1.00  0.00
ATOM     10  CA  GLY A  10      17.366  42.308  19.746  1.00  0.00
ATOM     11  CA  ALA A  11      15.981  42.045  23.277  1.00  0.00
ATOM     12  CA  GLY A  12      16.096  45.846  23.662  1.00 [... truncated]
Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      26.404  52.839  39.968  1.00  0.00
ATOM      2  CA  ARG A   2      27.389  49.887  37.790  1.00  0.00
ATOM      3  CA  ILE A   3      24.921  47.900  35.696  1.00  0.00
```

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ATOM      4  CA  ILE A   4      25.122  44.958  33.303  1.00  0.00
ATOM      5  CA  LEU A   5      22.436  44.472  30.647  1.00  0.00
ATOM      6  CA  LEU A   6      21.779  40.833  29.752  1.00  0.00
ATOM      7  CA  GLY A   7      19.622  39.236  27.078  1.00  0.00
ATOM      8  CA  ALA A   8      19.187  37.774  23.609  1.00  0.00
ATOM      9  CA  PRO A   9      19.955  39.586  20.331  1.00  0.00
ATOM     10  CA  GLY A  10      17.366  42.308  19.746  1.00  0.00
ATOM     11  CA  ALA A  11      15.981  42.045  23.277  1.00  0.00
ATOM     12  CA  GLY A  12      16.096  45.846  23.662  1.00 [... truncated]

Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      26.195  52.655  39.980  1.00  0.00
ATOM      2  CA  ARG A   2      27.445  49.862  37.675  1.00  0.00
ATOM      3  CA  ILE A   3      24.890  47.925  35.636  1.00  0.00
ATOM      4  CA  ILE A   4      25.249  44.923  33.275  1.00  0.00
ATOM      5  CA  LEU A   5      22.477  44.539  30.628  1.00  0.00
ATOM      6  CA  LEU A   6      22.174  40.847  29.713  1.00  0.00
ATOM      7  CA  GLY A   7      19.859  39.392  26.991  1.00  0.00
ATOM      8  CA  ALA A   8      19.284  37.781  23.510  1.00  0.00
ATOM      9  CA  PRO A   9      20.069  39.442  20.183  1.00  0.00
ATOM     10  CA  VAL A  10      17.852  42.616  19.704  1.00  0.00
ATOM     11  CA  ALA A  11      16.176  42.333  23.063  1.00  0.00
ATOM     12  CA  GLY A  12      16.235  46.140  23.333  1.00 [... truncated]
Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      26.195  52.655  39.980  1.00  0.00
ATOM      2  CA  ARG A   2      27.445  49.862  37.675  1.00  0.00
ATOM      3  CA  ILE A   3      24.890  47.925  35.636  1.00  0.00
ATOM      4  CA  ILE A   4      25.249  44.923  33.275  1.00  0.00
ATOM      5  CA  LEU A   5      22.477  44.539  30.628  1.00  0.00
ATOM      6  CA  LEU A   6      22.174  40.847  29.713  1.00  0.00
ATOM      7  CA  GLY A   7      19.859  39.392  26.991  1.00  0.00
ATOM      8  CA  ALA A   8      19.284  37.781  23.510  1.00  0.00
ATOM      9  CA  PRO A   9      20.069  39.442  20.183  1.00  0.00
ATOM     10  CA  VAL A  10      17.852  42.616  19.704  1.00  0.00
ATOM     11  CA  ALA A  11      16.176  42.333  23.063  1.00  0.00
ATOM     12  CA  GLY A  12      16.235  46.140  23.333  1.00 [... truncated]

Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      26.255  53.056  39.581  1.00  0.00
ATOM      2  CA  ARG A   2      27.451  50.025  37.599  1.00  0.00
ATOM      3  CA  ILE A   3      24.949  48.038  35.521  1.00  0.00
ATOM      4  CA  ILE A   4      25.077  44.956  33.303  1.00  0.00
```

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ATOM      5  CA   LEU A   5      22.336  44.433  30.718  1.00  0.00
ATOM      6  CA   LEU A   6      21.768  40.768  29.796  1.00  0.00
ATOM      7  CA   GLY A   7      19.632  39.610  26.899  1.00  0.00
ATOM      8  CA   ALA A   8      19.510  37.951  23.524  1.00  0.00
ATOM      9  CA   PRO A   9      20.522  39.822  20.371  1.00  0.00
ATOM     10  CA   GLY A  10      17.751  42.291  19.630  1.00  0.00
ATOM     11  CA   ALA A  11      16.234  42.203  23.127  1.00  0.00
ATOM     12  CA   GLY A  12      16.462  45.999  23.403  1.00 [... truncated]
Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA   MET A   1      26.255  53.056  39.581  1.00  0.00
ATOM      2  CA   ARG A   2      27.451  50.025  37.599  1.00  0.00
ATOM      3  CA   ILE A   3      24.949  48.038  35.521  1.00  0.00
ATOM      4  CA   ILE A   4      25.077  44.956  33.303  1.00  0.00
ATOM      5  CA   LEU A   5      22.336  44.433  30.718  1.00  0.00
ATOM      6  CA   LEU A   6      21.768  40.768  29.796  1.00  0.00
ATOM      7  CA   GLY A   7      19.632  39.610  26.899  1.00  0.00
ATOM      8  CA   ALA A   8      19.510  37.951  23.524  1.00  0.00
ATOM      9  CA   PRO A   9      20.522  39.822  20.371  1.00  0.00
ATOM     10  CA   GLY A  10      17.751  42.291  19.630  1.00  0.00
ATOM     11  CA   ALA A  11      16.234  42.203  23.127  1.00  0.00
ATOM     12  CA   GLY A  12      16.462  45.999  23.403  1.00 [... truncated]

Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA   MET A   1      26.688  52.893  40.331  1.00  0.00
ATOM      2  CA   ARG A   2      27.537  49.794  38.165  1.00  0.00
ATOM      3  CA   ILE A   3      24.935  47.859  36.170  1.00  0.00
ATOM      4  CA   ILE A   4      25.144  44.983  33.664  1.00  0.00
ATOM      5  CA   LEU A   5      22.360  44.828  31.030  1.00  0.00
ATOM      6  CA   LEU A   6      22.035  41.227  29.950  1.00  0.00
ATOM      7  CA   GLY A   7      19.331  39.980  27.611  1.00  0.00
ATOM      8  CA   ALA A   8      18.367  38.157  24.485  1.00  0.00
ATOM      9  CA   LEU A   9      19.146  39.512  21.086  1.00  0.00
ATOM     10  CA   VAL A  10      17.096  42.570  20.147  1.00  0.00
ATOM     11  CA   ALA A  11      15.347  42.647  23.576  1.00  0.00
ATOM     12  CA   GLY A  12      15.887  46.432  23.855  1.00 [... truncated]
Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA   MET A   1      26.688  52.893  40.331  1.00  0.00
ATOM      2  CA   ARG A   2      27.537  49.794  38.165  1.00  0.00
ATOM      3  CA   ILE A   3      24.935  47.859  36.170  1.00  0.00
ATOM      4  CA   ILE A   4      25.144  44.983  33.664  1.00  0.00
ATOM      5  CA   LEU A   5      22.360  44.828  31.030  1.00  0.00
ATOM      6  CA   LEU A   6      22.035  41.227  29.950  1.00  0.00
```

```
ATOM      7  CA  GLY A   7      19.331  39.980  27.611  1.00  0.00
ATOM      8  CA  ALA A   8      18.367  38.157  24.485  1.00  0.00
ATOM      9  CA  LEU A   9      19.146  39.512  21.086  1.00  0.00
ATOM     10  CA  VAL A  10      17.096  42.570  20.147  1.00  0.00
ATOM     11  CA  ALA A  11      15.347  42.647  23.576  1.00  0.00
ATOM     12  CA  GLY A  12      15.887  46.432  23.855  1.00 [... truncated]

Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      26.416  52.904  39.582  1.00  0.00
ATOM      2  CA  ARG A   2      27.690  49.867  37.720  1.00  0.00
ATOM      3  CA  ILE A   3      25.076  47.965  35.759  1.00  0.00
ATOM      4  CA  ILE A   4      25.077  44.962  33.429  1.00  0.00
ATOM      5  CA  LEU A   5      22.502  44.504  30.690  1.00  0.00
ATOM      6  CA  LEU A   6      22.169  40.802  29.941  1.00  0.00
ATOM      7  CA  GLY A   7      20.081  39.758  26.972  1.00  0.00
ATOM      8  CA  ALA A   8      19.954  38.450  23.416  1.00  0.00
ATOM      9  CA  PRO A   9      20.347  40.326  20.101  1.00  0.00
ATOM     10  CA  GLY A  10      17.481  42.738  19.406  1.00  0.00
ATOM     11  CA  ALA A  11      16.388  42.435  23.067  1.00  0.00
ATOM     12  CA  GLY A  12      16.214  46.192  23.635  1.00 [... truncated]
Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      26.416  52.904  39.582  1.00  0.00
ATOM      2  CA  ARG A   2      27.690  49.867  37.720  1.00  0.00
ATOM      3  CA  ILE A   3      25.076  47.965  35.759  1.00  0.00
ATOM      4  CA  ILE A   4      25.077  44.962  33.429  1.00  0.00
ATOM      5  CA  LEU A   5      22.502  44.504  30.690  1.00  0.00
ATOM      6  CA  LEU A   6      22.169  40.802  29.941  1.00  0.00
ATOM      7  CA  GLY A   7      20.081  39.758  26.972  1.00  0.00
ATOM      8  CA  ALA A   8      19.954  38.450  23.416  1.00  0.00
ATOM      9  CA  PRO A   9      20.347  40.326  20.101  1.00  0.00
ATOM     10  CA  GLY A  10      17.481  42.738  19.406  1.00  0.00
ATOM     11  CA  ALA A  11      16.388  42.435  23.067  1.00  0.00
ATOM     12  CA  GLY A  12      16.214  46.192  23.635  1.00 [... truncated]

Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      26.509  52.652  40.063  1.00  0.00
ATOM      2  CA  ARG A   2      27.933  49.470  38.495  1.00  0.00
ATOM      3  CA  ILE A   3      25.375  47.304  36.678  1.00  0.00
ATOM      4  CA  ILE A   4      25.363  44.770  33.835  1.00  0.00
ATOM      5  CA  LEU A   5      22.677  44.574  31.147  1.00  0.00
ATOM      6  CA  LEU A   6      21.986  41.012  30.001  1.00  0.00
ATOM      7  CA  GLY A   7      19.586  39.914  27.298  1.00  0.00
```

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ATOM      8  CA   ALA A   8      19.153  38.238  23.910  1.00  0.00
ATOM      9  CA   PRO A   9      20.068  40.265  20.800  1.00  0.00
ATOM     10  CA   GLY A  10      17.293  42.750  20.153  1.00  0.00
ATOM     11  CA   ALA A  11      15.858  42.616  23.677  1.00  0.00
ATOM     12  CA   GLY A  12      16.221  46.399  23.929  1.00 [... truncated]
Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA   MET A   1      26.509  52.652  40.063  1.00  0.00
ATOM      2  CA   ARG A   2      27.933  49.470  38.495  1.00  0.00
ATOM      3  CA   ILE A   3      25.375  47.304  36.678  1.00  0.00
ATOM      4  CA   ILE A   4      25.363  44.770  33.835  1.00  0.00
ATOM      5  CA   LEU A   5      22.677  44.574  31.147  1.00  0.00
ATOM      6  CA   LEU A   6      21.986  41.012  30.001  1.00  0.00
ATOM      7  CA   GLY A   7      19.586  39.914  27.298  1.00  0.00
ATOM      8  CA   ALA A   8      19.153  38.238  23.910  1.00  0.00
ATOM      9  CA   PRO A   9      20.068  40.265  20.800  1.00  0.00
ATOM     10  CA   GLY A  10      17.293  42.750  20.153  1.00  0.00
ATOM     11  CA   ALA A  11      15.858  42.616  23.677  1.00  0.00
ATOM     12  CA   GLY A  12      16.221  46.399  23.929  1.00 [... truncated]

Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA   MET A   1      26.440  52.814  40.254  1.00  0.00
ATOM      2  CA   ARG A   2      27.647  49.779  38.264  1.00  0.00
ATOM      3  CA   ILE A   3      25.155  47.915  36.008  1.00  0.00
ATOM      4  CA   ILE A   4      25.213  44.997  33.553  1.00  0.00
ATOM      5  CA   LEU A   5      22.527  44.398  30.822  1.00  0.00
ATOM      6  CA   LEU A   6      21.971  40.703  29.886  1.00  0.00
ATOM      7  CA   GLY A   7      19.647  39.398  27.143  1.00  0.00
ATOM      8  CA   ALA A   8      19.543  37.892  23.659  1.00  0.00
ATOM      9  CA   PRO A   9      20.726  40.066  20.708  1.00  0.00
ATOM     10  CA   GLY A  10      17.802  42.336  19.683  1.00  0.00
ATOM     11  CA   ALA A  11      16.165  42.309  23.102  1.00  0.00
ATOM     12  CA   GLY A  12      16.496  46.114  23.267  1.00 [... truncated]
Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA   MET A   1      26.440  52.814  40.254  1.00  0.00
ATOM      2  CA   ARG A   2      27.647  49.779  38.264  1.00  0.00
ATOM      3  CA   ILE A   3      25.155  47.915  36.008  1.00  0.00
ATOM      4  CA   ILE A   4      25.213  44.997  33.553  1.00  0.00
ATOM      5  CA   LEU A   5      22.527  44.398  30.822  1.00  0.00
ATOM      6  CA   LEU A   6      21.971  40.703  29.886  1.00  0.00
ATOM      7  CA   GLY A   7      19.647  39.398  27.143  1.00  0.00
ATOM      8  CA   ALA A   8      19.543  37.892  23.659  1.00  0.00
ATOM      9  CA   PRO A   9      20.726  40.066  20.708  1.00  0.00
```

```
ATOM     10  CA  GLY A  10      17.802  42.336  19.683  1.00  0.00
ATOM     11  CA  ALA A  11      16.165  42.309  23.102  1.00  0.00
ATOM     12  CA  GLY A  12      16.496  46.114  23.267  1.00 [... truncated]

Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      26.044  53.296  39.767  1.00  0.00
ATOM      2  CA  ARG A   2      27.287  50.263  37.778  1.00  0.00
ATOM      3  CA  ILE A   3      24.861  48.214  35.714  1.00  0.00
ATOM      4  CA  ILE A   4      25.048  45.169  33.423  1.00  0.00
ATOM      5  CA  LEU A   5      22.350  44.579  30.796  1.00  0.00
ATOM      6  CA  LEU A   6      21.767  40.988  29.677  1.00  0.00
ATOM      7  CA  GLY A   7      19.513  39.595  26.935  1.00  0.00
ATOM      8  CA  ALA A   8      19.433  38.203  23.388  1.00  0.00
ATOM      9  CA  PRO A   9      20.791  40.394  20.584  1.00  0.00
ATOM     10  CA  GLY A  10      17.804  42.577  19.665  1.00  0.00
ATOM     11  CA  ALA A  11      16.172  42.234  23.098  1.00  0.00
ATOM     12  CA  GLY A  12      16.182  46.013  23.621  1.00 [... truncated]
Warning in basename(x): expanded path length 17340 would be too long for
ATOM      1  CA  MET A   1      26.044  53.296  39.767  1.00  0.00
ATOM      2  CA  ARG A   2      27.287  50.263  37.778  1.00  0.00
ATOM      3  CA  ILE A   3      24.861  48.214  35.714  1.00  0.00
ATOM      4  CA  ILE A   4      25.048  45.169  33.423  1.00  0.00
ATOM      5  CA  LEU A   5      22.350  44.579  30.796  1.00  0.00
ATOM      6  CA  LEU A   6      21.767  40.988  29.677  1.00  0.00
ATOM      7  CA  GLY A   7      19.513  39.595  26.935  1.00  0.00
ATOM      8  CA  ALA A   8      19.433  38.203  23.388  1.00  0.00
ATOM      9  CA  PRO A   9      20.791  40.394  20.584  1.00  0.00
ATOM     10  CA  GLY A  10      17.804  42.577  19.665  1.00  0.00
ATOM     11  CA  ALA A  11      16.172  42.234  23.098  1.00  0.00
ATOM     12  CA  GLY A  12      16.182  46.013  23.621  1.00 [... truncated]

Warning in basename(x): expanded path length 16530 would be too long for
ATOM      1  CA  MSE A   1      27.296  51.234  40.972  1.00  0.00
ATOM      2  CA  ARG A   2      28.866  48.181  39.398  1.00  0.00
ATOM      3  CA  LEU A   3      26.650  47.145  36.447  1.00  0.00
ATOM      4  CA  ILE A   4      26.208  44.795  33.580  1.00  0.00
ATOM      5  CA  LEU A   5      23.737  46.288  31.039  1.00  0.00
ATOM      6  CA  LEU A   6      21.956  43.913  28.661  1.00  0.00
ATOM      7  CA  GLY A   7      19.712  45.044  25.774  1.00  0.00
ATOM      8  CA  ALA A   8      18.385  43.582  22.563  1.00  0.00
ATOM      9  CA  PRO A   9      19.441  45.284  19.324  1.00  0.00
ATOM     10  CA  GLY A  10      16.888  47.982  18.502  1.00  0.00
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ATOM     11  CA  ALA A  11      15.932  48.211  22.210  1.00  0.00
ATOM     12  CA  GLY A  12      17.830  51.535  22.520  1.00 [... truncated]
Warning in basename(x): expanded path length 16530 would be too long for
ATOM      1  CA  MSE A   1      27.296  51.234  40.972  1.00  0.00
ATOM      2  CA  ARG A   2      28.866  48.181  39.398  1.00  0.00
ATOM      3  CA  LEU A   3      26.650  47.145  36.447  1.00  0.00
ATOM      4  CA  ILE A   4      26.208  44.795  33.580  1.00  0.00
ATOM      5  CA  LEU A   5      23.737  46.288  31.039  1.00  0.00
ATOM      6  CA  LEU A   6      21.956  43.913  28.661  1.00  0.00
ATOM      7  CA  GLY A   7      19.712  45.044  25.774  1.00  0.00
ATOM      8  CA  ALA A   8      18.385  43.582  22.563  1.00  0.00
ATOM      9  CA  PRO A   9      19.441  45.284  19.324  1.00  0.00
ATOM     10  CA  GLY A  10      16.888  47.982  18.502  1.00  0.00
ATOM     11  CA  ALA A  11      15.932  48.211  22.210  1.00  0.00
ATOM     12  CA  GLY A  12      17.830  51.535  22.520  1.00 [... truncated]

Warning in basename(x): expanded path length 18393 would be too long for
ATOM      1  CA  THR A  -9      35.663  71.205  56.996  1.00  0.00
ATOM      2  CA  GLU A  -8      35.737  71.157  53.202  1.00  0.00
ATOM      3  CA  ASN A  -7      36.301  67.381  53.287  1.00  0.00
ATOM      4  CA  LEU A  -6      32.661  66.524  52.601  1.00  0.00
ATOM      5  CA  TYR A  -5      32.856  64.205  49.590  1.00  0.00
ATOM      6  CA  PHE A  -4      31.721  60.758  50.725  1.00  0.00
ATOM      7  CA  GLN A  -3      32.446  57.848  48.386  1.00  0.00
ATOM      8  CA  SER A  -2      29.249  55.763  48.429  1.00  0.00
ATOM      9  CA  ASN A  -1      29.151  51.987  47.903  1.00  0.00
ATOM     10  CA  ALA A   0      30.115  50.890  44.392  1.00  0.00
ATOM     11  CA  MSE A   1      27.223  50.214  42.054  1.00  0.00
ATOM     12  CA  ARG A   2      28.360  47.548  39.576  1.00 [... truncated]
Warning in basename(x): expanded path length 18393 would be too long for
ATOM      1  CA  THR A  -9      35.663  71.205  56.996  1.00  0.00
ATOM      2  CA  GLU A  -8      35.737  71.157  53.202  1.00  0.00
ATOM      3  CA  ASN A  -7      36.301  67.381  53.287  1.00  0.00
ATOM      4  CA  LEU A  -6      32.661  66.524  52.601  1.00  0.00
ATOM      5  CA  TYR A  -5      32.856  64.205  49.590  1.00  0.00
ATOM      6  CA  PHE A  -4      31.721  60.758  50.725  1.00  0.00
ATOM      7  CA  GLN A  -3      32.446  57.848  48.386  1.00  0.00
ATOM      8  CA  SER A  -2      29.249  55.763  48.429  1.00  0.00
ATOM      9  CA  ASN A  -1      29.151  51.987  47.903  1.00  0.00
ATOM     10  CA  ALA A   0      30.115  50.890  44.392  1.00  0.00
ATOM     11  CA  MSE A   1      27.223  50.214  42.054  1.00  0.00
ATOM     12  CA  ARG A   2      28.360  47.548  39.576  1.00 [... truncated]
```

```r
# Vector containing PDB database codes
ids <- basename.pdb(pdbs$id)

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```r
anno
```

```
       structureId chainId macromoleculeType chainLength experimentalTechnique
1AKE_A        1AKE       A           Protein         214                    X-
ray
```

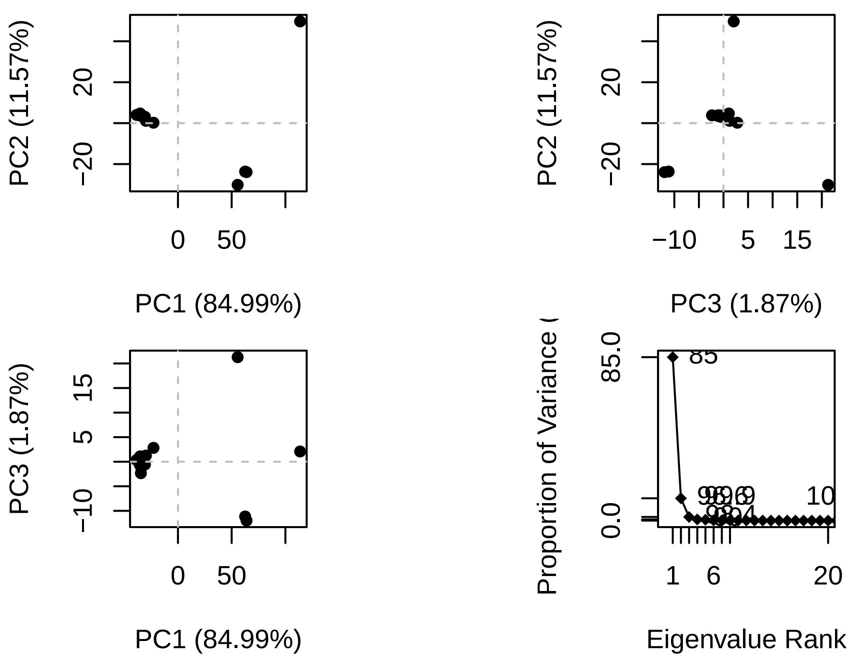
```
6S36_A        6S36       A          Protein        214                X-
ray
6RZE_A        6RZE       A          Protein        214                X-
ray
3HPR_A        3HPR       A          Protein        214                X-
ray
1E4V_A        1E4V       A          Protein        214                X-
ray
5EJE_A        5EJE       A          Protein        214                X-
ray
1E4Y_A        1E4Y       A          Protein        214                X-
ray
3X2S_A        3X2S       A          Protein        214                X-
ray
6HAP_A        6HAP       A          Protein        214                X-
ray
6HAM_A        6HAM       A          Protein        214                X-
ray
4K46_A        4K46       A          Protein        214                X-
ray
3GMT_A        3GMT       A          Protein        230                X-
ray
4PZL_A        4PZL       A          Protein        242                X-
ray
       resolution       scopDomain                                      pfam
1AKE_A       2.00 Adenylate kinase Adenylate kinase, active site lid (ADK_lid)
6S36_A       1.60             <NA> Adenylate kinase, active site lid (ADK_lid)
6RZE_A       1.69             <NA> Adenylate kinase, active site lid (ADK_lid)
3HPR_A       2.00             <NA> Adenylate kinase, active site lid (ADK_lid)
1E4V_A       1.85 Adenylate kinase Adenylate kinase, active site lid (ADK_lid)
5EJE_A       1.90             <NA> Adenylate kinase, active site lid (ADK_lid)
1E4Y_A       1.85 Adenylate kinase Adenylate kinase, active site lid (ADK_lid)
3X2S_A       2.80             <NA>                                      <NA>
6HAP_A       2.70             <NA> Adenylate kinase, active site lid (ADK_lid)
6HAM_A       2.55             <NA> Adenylate kinase, active site lid (ADK_lid)
4K46_A       2.01             <NA> Adenylate kinase, active site lid (ADK_lid)
3GMT_A       2.10             <NA>                                      <NA>
4PZL_A       2.10             <NA>                     Adenylate kinase (ADK)
              ligandId
1AKE_A             AP5
6S36_A CL (3),NA,MG (2)
6RZE_A    NA (3),CL (2)
3HPR_A             AP5
```

```
1E4V_A            AP5
5EJE_A         AP5,CO
1E4Y_A            AP5
3X2S_A   JPY (2),AP5,MG
6HAP_A            AP5
6HAM_A            AP5
4K46_A    ADP,AMP,PO4
3GMT_A        SO4 (2)
4PZL_A     CA,FMT,GOL
                                                                  ligandName
1AKE_A                                                 BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
6S36_A                                CHLORIDE ION (3),SODIUM ION,MAGNESIUM ION (2)
6RZE_A                                            SODIUM ION (3),CHLORIDE ION (2)
3HPR_A                                                 BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
1E4V_A                                                 BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
5EJE_A                             BIS(ADENOSINE)-5'-PENTAPHOSPHATE,COBALT (II) ION
1E4Y_A                                                 BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
3X2S_A N-(pyren-1-ylmethyl)acetamide (2),BIS(ADENOSINE)-5'-PENTAPHOSPHATE,MAGNESIUM ION
6HAP_A                                                 BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
6HAM_A                                                 BIS(ADENOSINE)-5'-
PENTAPHOSPHATE
4K46_A             ADENOSINE-5'-DIPHOSPHATE,ADENOSINE MONOPHOSPHATE,PHOSPHATE ION
3GMT_A                                                             SULFATE ION (2)
4PZL_A                                         CALCIUM ION,FORMIC ACID,GLYCEROL
                                                source
1AKE_A                                Escherichia coli
6S36_A                                Escherichia coli
6RZE_A                                Escherichia coli
3HPR_A                          Escherichia coli K-12
1E4V_A                                Escherichia coli
5EJE_A           Escherichia coli O139:H28 str. E24377A
1E4Y_A                                Escherichia coli
3X2S_A         Escherichia coli str. K-12 substr. MDS42
6HAP_A           Escherichia coli O139:H28 str. E24377A
6HAM_A                          Escherichia coli K-12
4K46_A                      Photobacterium profundum
3GMT_A                 Burkholderia pseudomallei 1710b
4PZL_A Francisella tularensis subsp. tularensis SCHU S4
```

```
1AKE_A STRUCTURE OF THE COMPLEX BETWEEN ADENYLATE KINASE FROM ESCHERICHIA COLI AND THE INHIBI
6S36_A
6RZE_A
3HPR_A
1E4V_A
loop
5EJE_A                                                                                 Crys
1E4Y_A
loop
3X2S_A
conjugated adenylate kinase
6HAP_A
6HAM_A
4K46_A
3GMT_A
4PZL_A                                                                             The crys
                                                     citation rObserved   rFree
1AKE_A              Muller, C.W., et al. J Mol Biology (1992)   0.19600      NA
6S36_A                  Rogne, P., et al. Biochemistry (2019)   0.16320 0.23560
6RZE_A                  Rogne, P., et al. Biochemistry (2019)   0.18650 0.23500
3HPR_A  Schrank, T.P., et al. Proc Natl Acad Sci U S A (2009)   0.21000 0.24320
1E4V_A                   Muller, C.W., et al. Proteins (1993)   0.19600      NA
5EJE_A  Kovermann, M., et al. Proc Natl Acad Sci U S A (2017)   0.18890 0.23580
1E4Y_A                   Muller, C.W., et al. Proteins (1993)   0.17800      NA
3X2S_A                Fujii, A., et al. Bioconjug Chem (2015)   0.20700 0.25600
6HAP_A               Kantaev, R., et al. J Phys Chem B (2018)   0.22630 0.27760
6HAM_A               Kantaev, R., et al. J Phys Chem B (2018)   0.20511 0.24325
4K46_A                   Cho, Y.-J., et al. To be published     0.17000 0.22290
3GMT_A Buchko, G.W., et al. Biochem Biophys Res Commun (2010)   0.23800 0.29500
4PZL_A                      Tan, K., et al. To be published     0.19360 0.23680
         rWork spaceGroup
1AKE_A 0.19600  P 21 2 21
6S36_A 0.15940    C 1 2 1
6RZE_A 0.18190    C 1 2 1
3HPR_A 0.20620  P 21 21 2
1E4V_A 0.19600  P 21 2 21
5EJE_A 0.18630  P 21 2 21
1E4Y_A 0.17800   P 1 21 1
3X2S_A 0.20700 P 21 21 21
6HAP_A 0.22370    I 2 2 2
6HAM_A 0.20311       P 43
4K46_A 0.16730 P 21 21 21
```

```
3GMT_A 0.23500    P 1 21 1
4PZL_A 0.19130        P 32
```

```
# Perform PCA
pc.xray <- pca(pdbs)
plot(pc.xray)
```

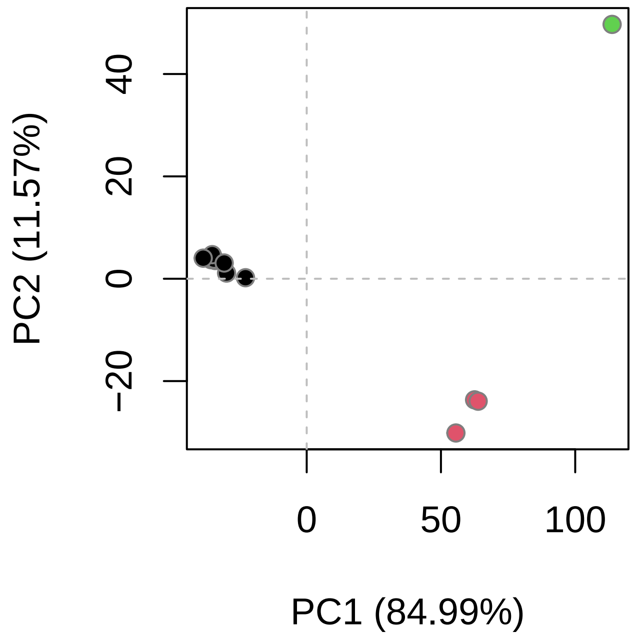



```
# Calculate RMSD
rd <- rmsd(pdbs)
```

```
Warning in rmsd(pdbs): No indices provided, using the 204 non NA positions
```

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```

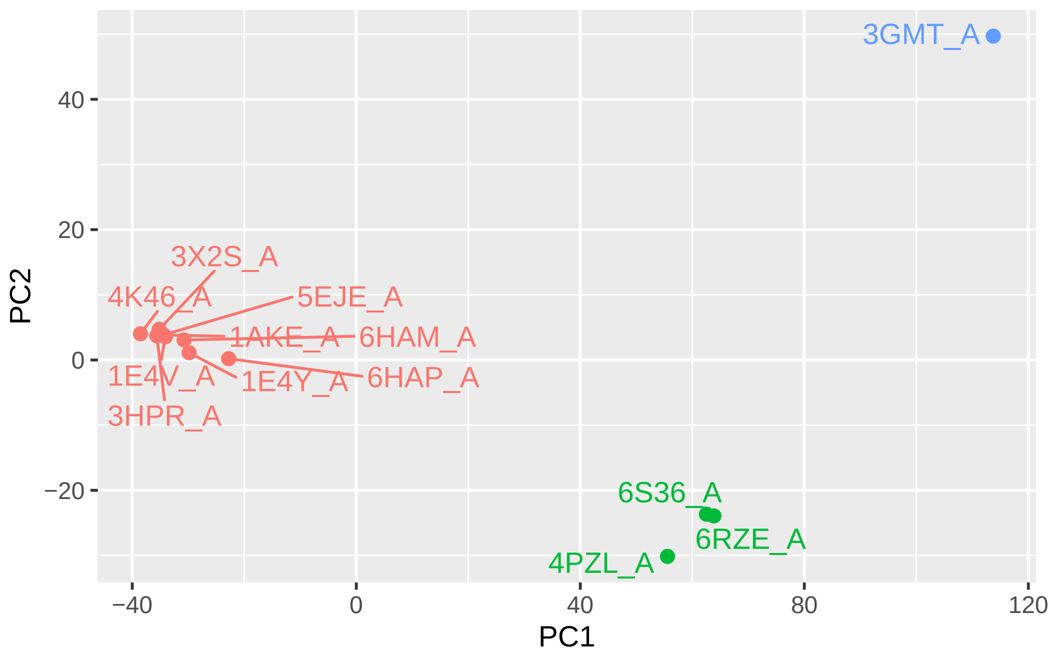



```
# Visualize first principal component
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                 PC2=pc.xray$z[,2],
                 col=as.factor(grps.rd),
                 ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```




```r
# NMA of all structures
modes <- nma(pdbs)
```

```
Details of Scheduled Calculation:
  ... 13 input structures
  ... storing 606 eigenvectors for each structure
  ... dimension of x$U.subspace: ( 612x606x13 )
  ... coordinate superposition prior to NM calculation
  ... aligned eigenvectors (gap containing positions removed)
  ... estimated memory usage of final 'eNMA' object: 36.9 Mb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  |
  |===========================================================           |  85%
  |
  |=================================================================     |  92%
  |
  |======================================================================| 100%
```
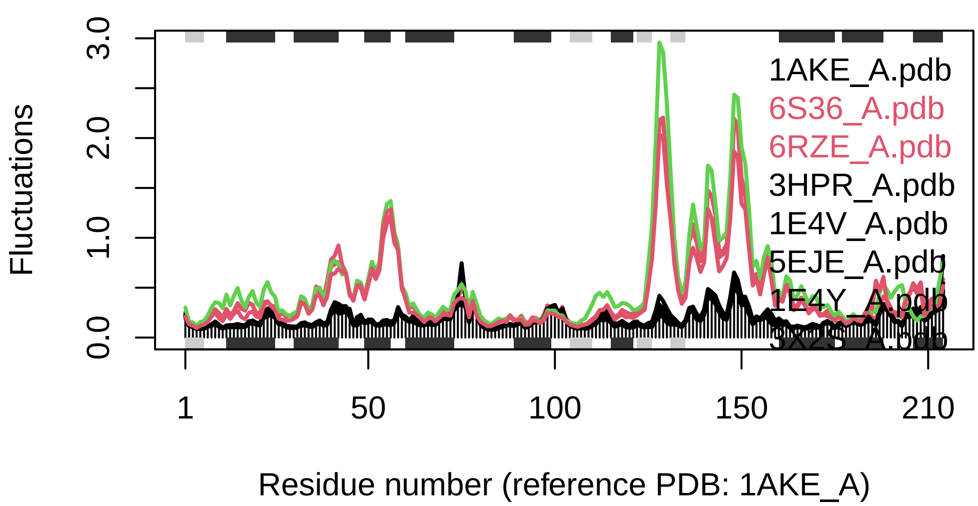

```r
plot(modes, pdbs, col=grps.rd)
```

```
Extracting SSE from pdbs$sse attribute
```



Q14. What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

> The black and colored lines are clearly different. The black lines show low and flat fluctuations indicating rigid, stable conformations, while the pink and green lines show much higher fluctuations, especially around residues 50 and 125–150. These regions differ most because Adk undergoes large conformational changes between open and closed states during its catalytic cycle, and the colored structures likely represent more open/flexible conformations while the black ones represent more closed/rigid conformations with ligand bound.